

MORE REDUCTIONS

Recall :

For any two languages $\mathcal{L}$ and $\mathcal{R}$ over alphabets Σ_1 and Σ_2 ,

If there is a total and computable function $\sigma: \Sigma_1^* \rightarrow \Sigma_2^*$ s.t.
for any $w \in \Sigma_1^*$, $w \in \mathcal{L}$ iff $\sigma(w) \in \mathcal{R}$, then,

we say that $\mathcal{L}$ reduces to $\mathcal{R}$ (denoted $\mathcal{L} \leq \mathcal{R}$), and

if $\mathcal{L}$ is (independently shown to be) undecidable, so is $\mathcal{R}$

$\mathcal{L} \leq \mathcal{R}$: $\mathcal{R}$ is at least as difficult (to decide) as $\mathcal{L}$

If $\mathcal{L}$ is undecidable, so is $\mathcal{R}$

Claim: $L_E = \{ \langle M \rangle \mid M \text{ is a TM and } M \text{ accepts } \epsilon \}$ is undecidable.

① Suppose L_E is decidable. There is a machine E deciding L_E .

② Want to show $L_{TM} \leq L_E$.

$\sigma(s) = \langle P_s \rangle$, where P_s does the following on input x :

(a) If s is of the form $\langle M \rangle \# \omega$ where M is a TM, proceed to (b).

(b) Run M on ω , accept x if M accepts ω . x is irrelevant!

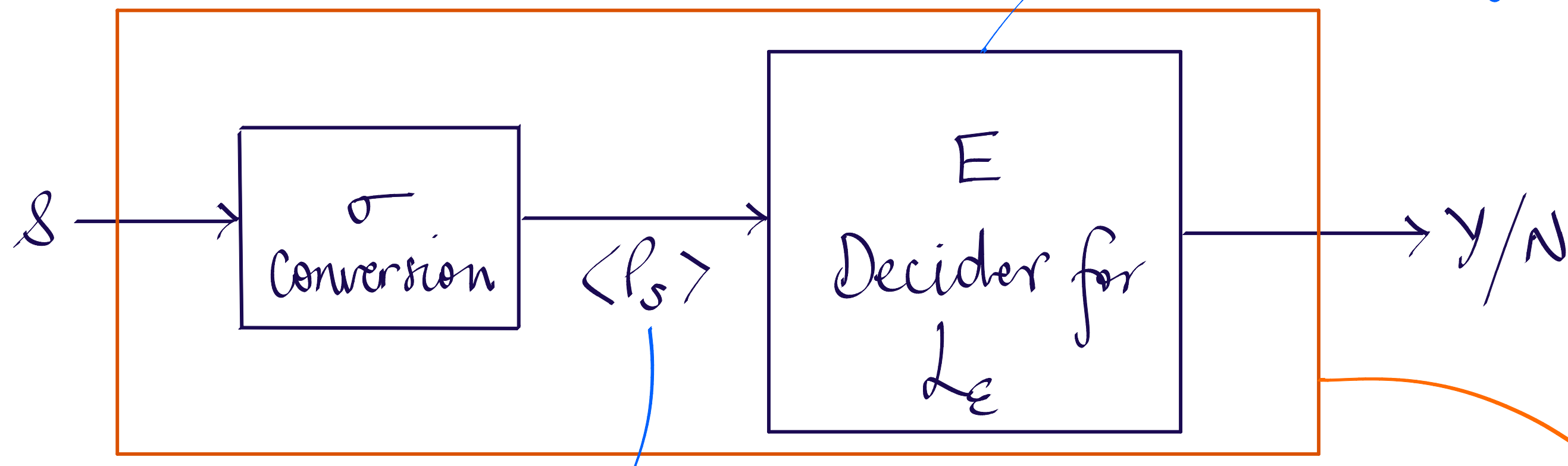
$$s \in L_{TM} \Rightarrow L(P_s) = \Sigma^* \Rightarrow \epsilon \in L(P_s) \Rightarrow \langle P_s \rangle \in L_E$$

$$s \notin L_{TM} \Rightarrow L(P_s) = \emptyset \Rightarrow \epsilon \notin L(P_s) \Rightarrow \langle P_s \rangle \notin L_E$$

$$\text{So, } s \in L_{TM} \Leftrightarrow \sigma(s) \in L_E.$$

Creating $\langle P_s \rangle$ is easy; it does not involve actually running P_s (or M)!

E accepts if P_s accepts ε ,
rejects if P_s does not accept ε



P_s accepts any string if s is of the form $\langle M \rangle \# w$,
where M is a TM and M accepts w ;
otherwise, P_s does not accept any string

This machine
decides L_{TM}

But L_{TM} is undecidable, so this yields a contradiction!
So our assumption about the decidability of L_E was wrong.
 $\therefore L_E$ is undecidable.

What about the following language?

$$L_s = \{ \langle M \rangle \mid M \text{ accepts } \langle M \rangle \}$$

Claim: $L_S = \{ \langle M \rangle \mid M \text{ accepts } \langle M \rangle \}$ is undecidable.

① Suppose L_S is decidable. There is a machine S deciding L_S .

② Want to show $L_{TM} \leq L_S$.

$\sigma(s) = \langle P_s \rangle$, where P_s does the following on input x :

(a) If s is of the form $\langle M \rangle \# \omega$ where M is a TM, proceed to (b).

(b) Run M on ω , accept x if M accepts ω .

$$s \in L_{TM} \Rightarrow L(P_s) = \Sigma^* \Rightarrow \langle P_s \rangle \in L(P_s) \Rightarrow \langle P_s \rangle \in L_S$$

$$s \notin L_{TM} \Rightarrow L(P_s) = \emptyset \Rightarrow \langle P_s \rangle \notin L(P_s) \Rightarrow \langle P_s \rangle \notin L_S$$

$$\text{So, } s \in L_{TM} \Leftrightarrow \sigma(s) \in L_S.$$

But L_{TM} is undecidable. So contradiction! L_S is undecidable.